Hexapod (Six-Legged Spider Robot)

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Abstract:

Six-legged robots or hexapods are famous now a days and a robotics engineer has built one of these kinds of robots once in their life. We decided to create a six-legged spider robot as a fun project for our Robotics course. Mechanical part includes the body of this robot which was designed in SolidWorks which included the design of coxa, femur, tibia (the three parts of hexapod leg) and a common base for all the legs. Electrical parts include servo motors, a microcontroller and servo drivers which were integrated on top of the base of the robot. We used forward kinematics for its kinematic model and its simulation was created on MATLAB. In the end this project worked and the robot could lay on the ground, stand up and walked.

I. INTRODUCTION

We have seen many kind of robots in science fiction movies and books and in the past people might have thinked that these robots will not e posible in the near future but in the present we can see many kinds of robots in labs and even in restaurants helping by becoming the waiters. We can see humanoid robots in labs that can jump and avoid obstacles and can even do back flips. In these kinds of robots one is a hexapod which can be defines as a six legged robot that can do various kind of actions.

There can be many different kinds of robots like the adaptive hexapod that can change its movement and sometimes geometry depending upon the envoronment [1] and also the legendary stewarts platform that is considered as a parallel manipultor which is like a 6 DOF platform that can move in three axis and can rotate in three planes. But for our project we made a spider robot that can move here and there. This robot has an advantage over other robots with its six legs this robot can stay in contact with the ground and is more stable then any other kind of robots. This also allows it to remain steady without any need of a stabalization sensor or complex balancing algorithm. Its design is based on insects and arthropods.

In this project we built our own six-legged spider walking robot from scratch. The main challenge was coordinating many movements with eighteen separate servo motors, three for each leg, to work together smoothly. To handle this complexity we used an ESP32-S3 microcontroller as the robot's brain and connected it to specialized servo driver modules. This setup managed the precise timing needed for smooth motion without overloading the main controller.

Our main goal was not to create a fully functioning robot with shiny features but to successfully combine the core systems. These include the mechanical structure, electronic control, and basic software, into a functional walking platform. We focused on making the robot walk stably using simple, predefined gait patterns. This project serves as a hands-on lesson in mechatronics by bringing together design, electronics, and programming principles.

II. SYSTEM DESIGN & METHODOLOGY

First of all lets define the system of this hexapod, it is a **Serial Manipulator** and has **3-DOF** for each leg. But for the whole body we will be designing it to **4-DOF** which will allow it to stay on the ground, stand up and then walk around. Lets do the forward kinematics of this robot first:

D-H Parameters

Joints	a	α	θ	d
1	0	pi/2	θ_1	0
2	L_1	0	θ_2	0
3	L_2	0	θ_3	0
4	L_3	0	0	0

Transformation Matrices

From 0 to 1:

$$ {}^{0}_{1}T = \begin{vmatrix} cos\theta_1 & 0 & sin\theta_1 & 0 \\ sin\theta_1 & 0 & -cos\theta_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} $$

From 1 to 2:

$$ {}^{1}_{2}T = \begin{vmatrix} cos\theta_2 & -sin\theta_3 & 0 & L_1cos\theta_2 \\ sin\theta_2 & cos\theta_2 & 0 & L_1sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} $$

From 2 to 3:

$$ {}^{2}_{3}T = \begin{vmatrix} cos\theta_3 & -sin\theta_3 & 0 & L_2cos\theta_3 \\ sin\theta_3 & cos\theta_3 & 0 & L_2sin\theta_3 \\ 0 & 0 & & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} $$

From 3 to 4:

$$ {}^{3}_{4}T = \begin{vmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & L_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} $$

Total Transformation:

$$T = \begin{vmatrix} cos\theta_1 * cos(\theta_2 + \theta_3) & -cos\theta_1 * sin(\theta_2 + \theta_3) & sin\theta_1 & cos\theta_1 * (L_1 cos\theta_2 + L_2 cos(\theta_2 + \theta_3) + L_3 cos(\theta_2 + \theta_3)) \\ sin\theta_1 * cos(\theta_2 + \theta_3) & sin\theta_1 * sin(\theta_2 + \theta_3) & cos\theta_1 & sin\theta_1 * (L_1 cos\theta_2 + L_2 cos(\theta_2 + \theta_3) + L_3 cos(\theta_2 + \theta_3)) \\ sin(\theta_2 + \theta_3) & cos(\theta_2 + \theta_3) & 0 & L_1 sin\theta_2 + L_2 sin(\theta_2 + \theta_3) + L_3 sin(\theta_2 + \theta_3) \\ 0 & 0 & 0 & 1 \end{vmatrix}$$

After finding out the transformation we wrote a simple code in matlab for plotting this robot and making its simulation. The image of MATLAB plot is given below:

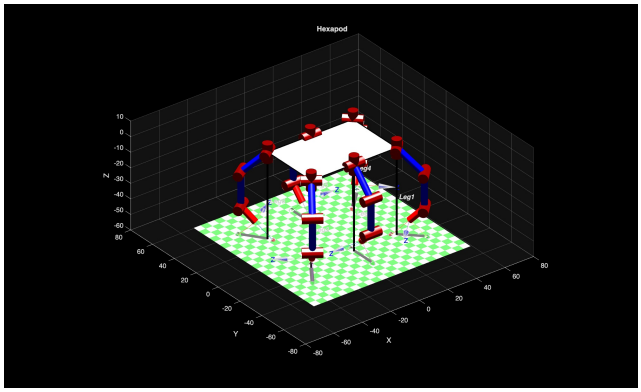

Figure 1 an image of a hexapod created in MATLAB

The rest of the design can be divided into three categories:

Mechanical Design – Focused on leg geometry and joint arrangement

Electronic System – Integrating the controller, servo drivers, power management, and wiring.

Software and Control – Responsible for gait generation, timing, and servo coordination.

III. MECHANICAL DESIGN

The robot's body was made using 3D printing in lightweight material PLA, with six identical legs symmetrically arranged. Each leg has three rotational joints, providing three degrees of freedom (3-DOF):

Coxa (Hip) Joint – Controls forward/backward leg swing.

Femur (Knee) Joint – Controls leg lifting.

Tibia (Ankle) Joint – Fine-tunes foot placement and stance.

The CAD model is given below:

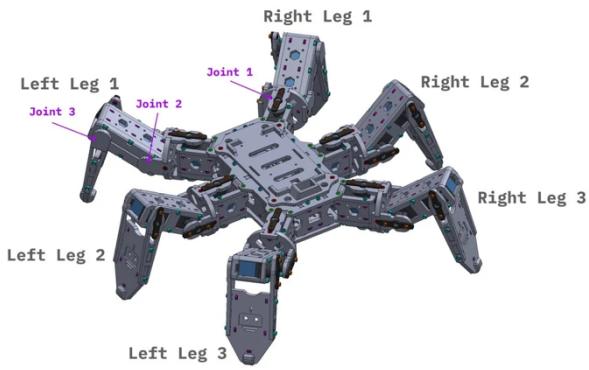

Figure 2 an image of CAD model of hexapod created in Solidworks

This configuration allows omnidirectional movement and adaptability to minor terrain variations. The structure was designed to keep the center of mass low and central, improving static stability.

IV. ELECTRONIC SYSTEM

Control Unit – ESP32-S3

The ESP32-S3 [2] was selected for its fast dual-core processor, sufficient GPIO pins, and built-in wireless features for future expansions. It runs the main control loop, calculating joint angles and sending commands to the servo drivers.

Actuation – Servo Motors

Eighteen standard hobby servo motors (e.g., MG90S) provide actuation. Each offers adequate torque for the robot's weight and speed suitable for stable walking [3].

Servo Drivers – PCA9685 Modules

Two 16-channel PCA9685 PWM drivers handle all servo signals. They communicate with the ESP32-S3 via I²C, offloading precise PWM generation and reducing processing overhead.

Power System

A dual-supply setup is used: a 5V/3A regulator for the electronics and a separate 6V/5A supply for the servos. Capacitors are placed near the drivers to mitigate voltage drops during sudden servo movements.

V. SOFTWARE & CONTROL LOGIC

The control software was developed in the Arduino IDE (C++). Key functions include:

Forward Kinematics (FK) – Converts joint angles to position in x, y, z components.

Gait Scheduler – Implements a tripod gait: three legs (alternating) lift while the other three support the body [4].

Communication Routine – Sends angle data via I²C to the PCA9685 drivers.

The gait timing and step height are parameterized, allowing easy tuning for different speeds and stability preferences.

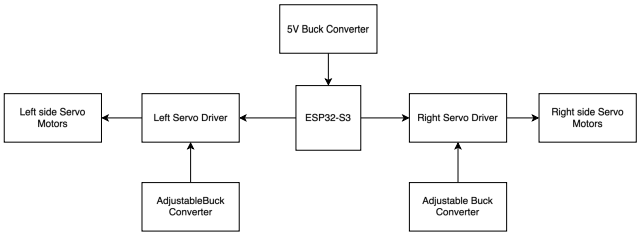

Figure 3 A block diagram showing all the connections in the hexapod for easier understanding.

VI. RESULTS & DISCUSSION

The hexapod successfully walked using a continuous tripod gait without tipping or desynchronization. The ESP32-S3 and dual-driver setup proved effective in managing all eighteen servos in real time. Mechanical rigidity and joint alignment were critical; slight misalignments caused noticeable gait deviations. Power consumption averaged 2.5A during motion, with no significant driver overheating.

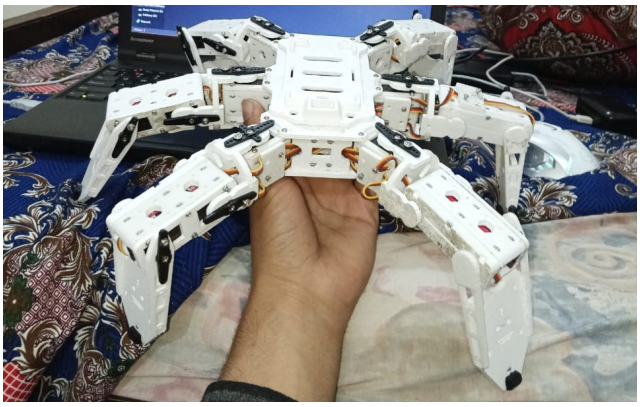

Figure 4 Image of working prototype of hexapod

Limitations observed:

- Limited payload capacity due to servo torque.
- Gait is preprogrammed; no real-time terrain adaptation.
- Wireless control and sensors not yet integrated.
- These limitations define clear paths for future work.

VII. CONCLUSION & FUTURE UPGRADES

This project successfully realized a functional hexapod robot using widely available components. The integration of the ESP32-S3 with PCA9685 drivers offers a scalable control architecture for multi-legged robots. The platform walks stably and can serve as an educational tool for mechatronics and robotics courses.

Future upgrades may include:

- Adding ultrasonic/IR sensors for obstacle detection.
- Implementing a Bluetooth smartphone remote control.
- Experimenting with adaptive gaits using an IMU for slope compensation.
- Designing a stronger, lighter chassis with 3D-printed parts.

VIII. REFERENCES

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